

Practice Midterm

Name and ID: _____

Instructions: No notes, books, calculators, cellphones, smartwatches, or supplies except pencils and pens allowed.

Academic Integrity: Write your name at the top of every page. Keep your work covered.

Suggestions: Don't spend too long on any one problem. Use the scratch paper.

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1. True or False? Give convincing reasons for your answers.

(a) If A is a 4×3 matrix then the system of equations $AX = b$ must be inconsistent for some vector b in $\mathbb{R}^4$.

(b) If A is a 3×4 matrix then the matrix equation $AX = 0$ must have infinitely many solutions X in $\mathbb{R}^4$.

(c) If A is a 10×5 matrix then the system of equations $AX = 0$ is consistent.

(d) If m vectors $v_1, v_2, \dots, v_m$ span $\mathbb{R}^4$ and they are linearly independent, then $m = 4$.

(e) Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Then T is one-to-one if in the standard matrix of T , every column has a pivot position.

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(f) The set $\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 1 \\ 3 \end{bmatrix} \right\}$ is linearly independent.

(g) A set of 5 vectors in $\mathbb{R}^4$ is always linearly dependent.

(h) The set $\left\{ \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \begin{bmatrix} 2 \\ 8 \end{bmatrix} \right\}$ is linearly independent.

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(i) The matrix $\begin{bmatrix} 1 & 5 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ is invertible.

(j) If A is a 3×3 matrix with 3 pivot positions, then A is invertible.

(k) A system $Ax = b$ is inconsistent if the augmented matrix $[A \mid b]$ has a pivot in the last column.

(l) Every homogeneous system $Ax = 0$ has at least one solution.

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(m) If A is a 4×3 matrix, the columns of A can span $\mathbb{R}^4$.

(n) The column space $Col(A)$ of a 5×3 matrix A is a subspace of $\mathbb{R}^3$.

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2. Solve the following system using an augmented matrix and row reduction. Write the solution in **parametric vector form**.

$$x_1 + 2x_2 + x_3 + 3x_4 = 1$$

$$2x_1 + 4x_2 + 3x_3 + 8x_4 = 4$$

$$3x_1 + 6x_2 + 4x_3 + 11x_4 = 5$$

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3. Find the inverse of matrix A using row reduction. Show all row operations clearly.

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 3 \\ 0 & 1 & 1 \end{bmatrix}$$

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4. Consider the matrix

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \end{bmatrix}.$$

(a) Solve the homogeneous equation $AX = 0$ and write your solution in the parametric vector form.

(b) Use part (a) to solve the non-homogeneous equation $AX = \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix}$.

Hint: Guess a *particular solution*.

(c) Is the linear transformation $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$, given by $T(X) = AX$, *onto*?

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5. Interpret the following linear transformation geometrically.

$$T(X) = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} X$$

6. $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the rotation by $5\pi/3$ in the counter-clockwise direction followed by a reflection about the line $x_2 = 0$. Find the standard matrix of T .

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7. Show that the map $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ given by

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2^x + 2^y - 2 \\ x + y \\ x - y \end{bmatrix},$$

is not a linear transformation.

8. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^4$ be a linear transformation such that

$$T \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{and} \quad T \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}.$$

Find the standard matrix of T . Then find $T \begin{bmatrix} 4 \\ 6 \end{bmatrix}$.

Scratch Paper